Principal Component Analysis and Galaxy Eigenspectra

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Principal Component Analysis (PCA) is currently one of the most powerful dimension reducing algorithms available. In this paper, we utilize Randomized PCA to compress galaxy spectra data harvested from the SDSS. We analyze the eigenspectra that were calculated and broaden the discussion to the use of PCA and other dimensionality reduction techniques in computational science.

1. INTRODUCTION

The Sloan Digital Sky Survey (SDSS) launched in 2000 and has since been one of the most ambitious and successful mappings of the sky to date. It has covered more than a quarter of the sky and has catalogued more than 930,000 galaxies and their respective properties. In this paper, we plan to analyse the spectra of a sample of galaxies from the SDSS. Spectra data is formed by observing the amount of flux observed at each wavelength of light coming from a galaxy. This data provides us a clear view of the material that makes up the galaxy. For example, when ionized hydrogen moves to a lower energy level, it emits light in a specific wavelength. These wavelengths with a large spike in flux are usually called emission lines. Conversely, wavelengths with a sharp decrease in flux are called absorption lines (theoretically because a material is absorbing a specific wavelength of light). These galactic fingerprints help provide researchers a clue as to what these galaxies are comprised of. Analysing this data, however, comes with challenges. First, spectra data spanning a large range of wavelengths for just a single galaxy is fairly large. Even for a sample of a couple thousand galaxies, the amount of data points can reach the millions. This makes performing any meaningful calculations difficult. There is a solution.

With the advent of the modern computer, physics and other similar fields have seen an incredible growth in capability. Suddenly, problems that were difficult to solve have become trivial thanks to a rapid increase in computational power. Furthermore, an explosion of new techniques, thanks to advances in computer science and machine learning, have equipped physicists with a plethora of tools for them to conduct complex analyses that previously would've been nearly impossible. One such tool, principal component analysis (PCA), will be the primary computational tool of this paper. PCA is a dimensionality reduction technique that can reduce the size of the data to only the most essential components. PCA need not be just a processing tool either as these principle components could also help illuminate hidden characteristics of the data, in this case the spectra data of the galaxies. In this paper, we plan to perform PCA on the spectra data of a sample of galaxies form the SDSS. The techniques used are heavily inspired by [7], [6], and [2].

In Section 2 we outline the data used and some of the pre-processing techniques, in Section 3 we go over technique of Principal Component Analysis, in Section 4 we perform an analysis of the data's eigenspectra, and in Section 5 we summarize the paper's results and extend it to the broader discussion of computational physics.

2. DATA

The dataset we use is comprised of 4000 galaxies and their flux values ranging from 3000 Angstroms to 8000 Angstroms (roughly 300nm to 800nm). This range of wavelengths provides us with a strong sample of possible emission lines like the Hα emission line which occurs at 656nm. Each galaxy has 1000 points of spectra data associated with it. This yields 4,000,000 total data points. A majority of the galaxy data has been retrieved from the SDSS.

Its important to note that the data from the SDSS is not perfect. There are at times gaps in the data since the SDSS can be limited in what it can observe. Working with this raw data can create an incomplete picture. In addition, it can create complications when performing calculations. Some pre-processing is required before we can begin using the data. The actual dataset we use is scrapped from the astroML python library. The author, Jake Vanderplas, uses techniques outlined in [7] for gap correction. In the paper, the authors actually use PCA (also called the Karhunen-Loeve Transform) to fill in missing gaps. This is because PCA as a tool is not only powerful for analysing essential components, but also providing predictions based on provided input (we discuss this more in detail in Section 3). In essence the data has been taken from the SDSS and had the first 20 eigenmodes calculated. Then using a process known as reconstruction, flux values are predicted for wavelengths where there are clear gaps. Its important to note that these flux values are still machine made and might not fully capture finer behaviors of the data. We can tune this process by including more eigenmodes in reconstruction, but we experience diminishing returns. PCA is itself a data reduction technique. Looking for these finer details is not needed as we plan on excluding them anyway when we conduct our analysis in Section 4. Finally, the data is

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normalized. From here we can begin discussion on PCA.

3. PRINCIPAL COMPONENT ANALYSIS

3.1. Calculating the Principal Components

The Karhunen-Loeve Transform (KL), or principal component analysis (PCA) as its commonly called, is an unsupervised machine learning technique designed for dimensionality reduction. Consider the weather: if we want to predict the weather we need to account for temperature, humidity, precipitation, air pressure, etc. All of these variables means a large number of dimensions that our data spans. Predicting the weather with such a large number of dimensions is computationally expensive and extremely difficult even for a computer. The philosophy behind PCA is to reduce the complexity of the data but retain key characteristics. We start with a data set of values and form a covariance matrix

$$\mathbf{C} = \begin{bmatrix} COV(x_1, x_1) & COV(x_1, x_2) & \dots & COV(x_1, x_n) \\ COV(x_2, x_1) & COV(x_2, x_2) & \dots & COV(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ COV(x_n, x_1) & COV(x_n, x_2) & \dots & COV(x_n, x_n) \end{bmatrix} \tag{1}$$

where

$$COV(x_n, x_m) = E[(x_n - E[x_n])(x_m - E[x_m])]. \tag{2}$$

It should be noted that when $n = m$, we simply have $VAR(x_n)$. Thus, we have the variance of each variable going along the diagonal of $\mathbf{C}$. The covariance matrix mathematically tells us how each variable is related to each other. Specifically, how much of the variance of the data is accounted for by each variable. We can now look to calculate the principal components. To do this, we will find the eigenvalues and eigenvectors of $\mathbf{C}$. We have

$$|C - \lambda \mathbf{I}| = 0 \tag{3}$$

and

$$\mathbf{C}\vec{v} = \lambda \mathbf{C} \tag{4}$$

where $\vec{v}$ are the eigenvectors and λ are the corresponding eigenvalues. We call the eigenvectors the principal components of the data. A principal component's corresponding eigenvalue represents how much variance of the data that the component accounts for. We order the principal components on the magnitude of their eigenvalues with the first principal component being the eigenvector with the largest eigenvalue. Therefore, the first principal component represents the largest share of variance within the data. We continue this process for all principal components we intend to calculate. PCA works because we're able to project our data from a higher dimension to a lower, much easier to work with dimension

$$\mathbf{X}_p = \mathbf{XR} \tag{5}$$

where $\mathbf{X}_p$ is our projected data, $\mathbf{X}$ is our original data, and $\mathbf{R}$ is a matrix whose columns are the eigenvectors.

3.2. Why Perform PCA

One of the primary benefits to PCA is that it is a dimensionality reduction tool. A data set with a large number of dimensions can easily be compressed to only those that account for the largest variance. Often, the number of principal components required to represent at least 90% of the data is orders of magnitude less than the total number of dimensions. Thus, the computational cost can be greatly reduced. Therein lies another benefit however.

Because the first couple of principal components can explain most of the variance within the data, then we can create predictions based on the relationships exposed by our principal components. This is the basis behind reconstruction. Specifically, with our basis set of eigenvectors, which are by definition linearly independent, we can mathematically predict a value for flux at a wavelength by forming a linear combination of principal components [7]. As mentioned previously in Section 2, we can never account for 100% of the variance in the data as that would require us to consider all dimensions (which is antithetical to the purpose of PCA).

Finally, we want to discuss the physical representation of the principal components themselves. While [7] endeavours to analyse these eigenspectra and provide physical context, it should be noted that the principal components (as a matter of fact quite often) don't describe any sort of physical reality. The eigenvectors are mathematically defined relative to the mean; as a result they can posses values that can seem nonsensical if taken at face value. Figure 1 shows how the eigenspectra don't resemble any form galaxy spectra despite being descendent of galaxy spectra data. Furthermore, while the eigenspectra are by definition orthonormal as to satisfy the condition of forming a basis set, they are not statistically orthogonal [2]. Variables within the data are not entirely uncorrelated. For example, we should expect to see similar metal absorption lines if we notice Ca H and K absorption lines since this indicates an older likely elliptical galaxy. One can construct a basis set that is in fact statistically orthogonal which would require satisfying the condition

$$\langle X_i X_j^* \rangle = \langle X_i^2 \rangle \delta_{ij} \tag{6}$$

where δ is the Kronecker delta [6]. However, that is beyond the scope of this paper. Because the eigenspectra

are not statistically orthogonal, they can't necessarily be studied independently of each other. While they don't possess any real physical reality, their effects on the data can still be analyzed.

3.3. Randomized PCA

While PCA itself is designed to reduce computational work load (among other reasons) as outlined in 3.2, for larger data sets with a lot of dimensions, performing PCA itself is intensive. From here we can turn to randomized dimensionality reduction. The central idea is that instead of solving for our eigenvectors, we can randomly chose a matrix $\mathbf{R}_r$ that is capable of projecting $\mathbf{X}$ to lower dimensions

$$\mathbf{X}_{rp} = \mathbf{R}_r \mathbf{X} \tag{7}$$

where $\mathbf{X}_{rp}$ is our random projection [1]. This technique is made possible by the Johnson-Lindenstrauss lemma which states that if points in a vector space are projected onto a subspace of high enough dimensions, then the distances between points are preserved. In context, we can randomly project our data to a lower dimension and as long as the dimension is large enough, then the behavior of our data still remains. In order to be a true projection however, the column vectors of $\mathbf{R}_r$ must be orthogonal. Luckily, because the number of near orthogonal vectors within a space are much larger than the number of truly orthogonal vectors [5], randomly choosing vectors is sufficient to produce a matrix such that

$$\mathbf{R}_r^T \mathbf{R}_r \approx \mathbf{I}. \tag{8}$$

And so, for the purposes of this paper, we will be employing randomized PCA.

4. EIGENSPECTRA

We look to study the mean plus the following 4 principal components of our data set. The bulk computation for the principal components was performed by the python library scikit-learn. Specifically, we utilized the `PCA` class from their `sklearn.decomposition` module. The class is capable of calculating the nth principal component and its corresponding eigenvector. For computational speed, we employ the `RandomizedPCA` method from the class. Figure 1 showcases the plots of the principal components. Figure 2 shows a sample of 5 spectra randomly taken for reference.

4.1. First Principal Component

Following in [7], our first principal component greatly resembles the mean. It logically follows that the first principal component mathematically represents the broadest trends that exist within the data. Stronger emission and spectral lines are omitted however. For example, the emission line that exists around 3800 Angstrom is noticeably absent from the first eigenspectrum.

4.2. Second Principal Component

The second principal component appears to be an outlier. There is a wide spike between 3000 and 3600 Angstroms before flattening. Again, the components themselves need not physically represent anything. However, from viewing Figure 2, we can see there is a much greater amount of noise within this range for the 5 randomly sampled spectra than anywhere else within the plots. The second eigenspectrum appears to represent the variance that comes from this noise.

4.3. Further Components

The third and fourth principal components are relatively flat with very little variation. There is the occasional spike and dip which could represent absorption and emission lines. The dip at 3800 angstroms is very reminiscent of metal absorption lines. However, in terms of variance we can see how by the third eigenspectrum, most of the variance in the data has already been accounted for.

5. CONCLUSION

We were able to successfully calculate the principal components of the data and perform an analysis of the resulting eigenspectrum. PCA remains as one of the most powerful computational techniques available. As detailed in Section 3.2, there are a multitude of reasons as to why one would want to perform PCA. We look to broaden the discussion beyond galaxy spectra and more towards general usage.

First, PCA is used all over the world in a multitude of different fields. For example, in medicine, PCA is being used to help denoise and improve MRI imaging [3]. In addition, PCA as a tool is constantly evolving and being reimagined. In image processing, researchers have developed a new form of PCA called Tensorial PCA (or TPCA) that allows them to compress compound pixel data [4]. With such widespread use and constant innovation, its a wonder why PCA isn't used in nearly every analysis.

For one, PCA is a linear process. It is an essential assumption that the data undergoing PCA is behaves linearly. Theoretically, one could still perform PCA on nonlinear data provided they have enough principle components, however, this defeats the purpose of using PCA in

the first place. The computational cost of doing such an analysis outweighs any benefit from compression. Such a process is better suited by Local Linear Embedding (LLE). LLE works by zooming in close enough such that the local geometry can be approximated linearly. From here, the process of dimension reduction is similar to PCA. Furthermore, as discussed in Section 4, the principal components are not statistically independent. Thus, we aren't guaranteed to extrapolate any physical meaning by analyzing the components themselves. For such an analysis, we look to Independent Component Analysis (ICA).

Regardless of these caveats, PCA still remains as one of the most prominent tools on the market. In the future, we hope to pair our results from this analysis with another machine learning algorithm. In a similar fashion, neural networks have begun to dominate the field of physics and computer science. Restricted Boltzmann Machines (RBM) are neural networks that work similarly to PCA in that they help reduce the dimensionality of data. We would look to compare the precision, speed, etc. of an unsupervised learning algorithm like PCA to a powerful neural network like RBM on a sample of galaxy spectra data.

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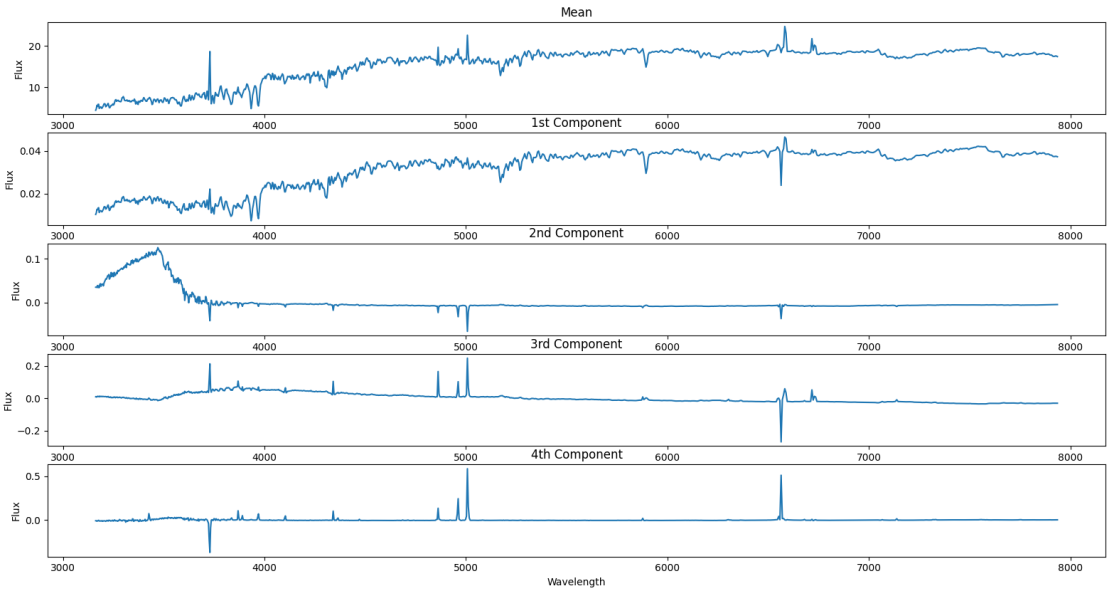

FIG. 1: The mean plus first four principle components plotted.

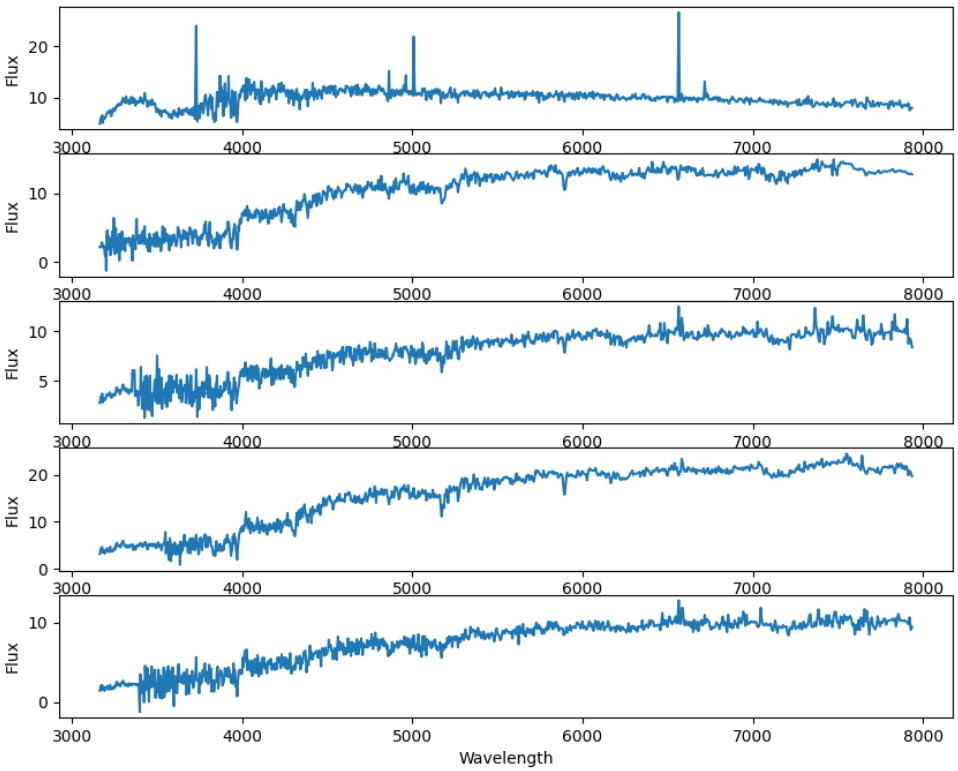

FIG. 2: A sample of 5 randomly chosen spectra from the dataset.